

Abhishek Tiwari

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Summary — Full-Stack Software Engineer with 2.5+ years of experience building scalable SaaS applications using React, Redux, Tailwind, FastAPI, Django, Flask, and Spring Boot. Skilled in designing modern APIs, cloud-native backend systems, and data-intensive pipelines across AWS, PostgreSQL, Redis, and RabbitMQ. Delivered AI-powered RAG search, optimized API performance, automated CI/CD, and built multi-tenant platforms in Agile environments.

Skills

Languages	C++, Python, JavaScript, TypeScript, SQL, HTML, CSS, GO	Cloud & DevOps	Boot, LangChain, Pandas, NumPy, Node.js, LLVM AWS (RDS, ECS, S3), GCP (Cloud Run, Cloud SQL, IAM), Docker, GitLab CI/CD, Airflow, ELK
Developer Tools	VS Code (IDE), Git, GitHub, RabbitMQ, MongoDB, MySQL	Frontend	React, Redux, Angular
Frameworks	FastAPI, Flask, Django, Express.js, Spring		

Experience

Crystaltech Services Private Limited

Jan 2024 – Present

Full Stack Software Engineer

- Led the launch of a **cloud-based SaaS platform** with **300+ MAUs in 5 months**, implementing **PostgreSQL multi-tenancy** to enable 5 state departments and increase adoption **5x**.
- Refactored front-end with React, Redux, Tailwind, optimized FastAPI APIs with Redis (reducing latency by 3s), developed RAG-based GPT-4/LangChain search for 450+ PDFs, and automated 600+ unit tests via GitLab CI.
- Automated data migrations via Airflow, standardizing deployments and accelerating release cycles.
- Contributed to GCP migration PoCs (Cloud Run, Cloud SQL, IAM) and assisted in building prototype dashboards using
- **Tech Stack:** React, FastAPI, Redis, PostgreSQL, LangChain/GPT-4, Airflow, GitLab CI, AWS/GCP

Crystaltech Services Private Limited

July 2023 – Dec 2023

Associate Software Engineer

- Built **Django REST APIs** and data models for **700K+ responses**, enabling nationwide telecom analysis.
- Centralized multi-department data into **SQL Server** powering 25 **Tableau dashboards**, implemented **Redis middleware** for API idempotency, and integrated **ELK stack** for auditability and real-time monitoring.
- Maintained data provenance using SQL Server triggers and audit tables for complete historical tracking.
- Supported automation of data migrations via Airflow, standardizing workflows and reducing manual effort
- **Tech Stack:** Django REST, SQL Server, Redis, ELK Stack, Airflow, Tableau

Education

University of Allahabad

Aug 2020 – July 2023

CGPA: 7.2

Bachelor of Science in Mathematics

Minors: Statistics ; Computer Programming ; Applied Mathematics ; Mathematical Modelling ; Differential Equations

Projects

Heart Disease Prediction

- Developed a heart disease prediction model using **XG-Boost** and **SVM** algorithms on a dataset of **300+** medical records.
- Achieved **99%** recall and **97%** accuracy, identifying high-risk patients with minimal false negatives.
- Deployed the model using **Flask** with an interactive **JavaScript** frontend, handling 100+ real-time predictions/day
- **Tech Stack:** Machine Learning, Python, Flask, JavaScript

Phishshield Mails

- Architected a secure email management system using APIs, enhancing efficiency and reducing API latency by **30%**.
- Developed and deployed an AI-driven email classifier, reducing manual email sorting efforts by **45%** and enhancing phishing detection accuracy.
- Optimized a React-based web app interface, processing **100+** mails simultaneously, boosting engagement by **25%** through an enhanced user experience.
- Built an Angular prototype of an email analytics dashboard to explore component architecture, routing, and service-based **API** integration for future migration.
- **Tech Stack:** ReactJS, JavaScript, CSS, Generative AI, Google Auth, Web Analytics

AI Job Tracker

- Developed a **HATEOAS-compliant REST API** for astronomers to record and share observations.
- Deployed a **fault-tolerant AWS ECS system**, implemented caching and asynchronous email alerts to improve performance.
- Leveraged **Redis and RabbitMQ** for optimized caching and messaging, ensuring smooth scaling and reliability.
- Built cloud-native PoC on GCP to deploy microservice and evaluate portability, scaling, security, and cost.
- **Tech Stack:** Java, Spring Boot, PostgreSQL, AWS, Redis, RabbitMQ